

Solution: $p: 3x+2$ is even $q: x+5$ is odd $r: x^2$ is even.

We will prove p, q and r are equivalent by showing that each of $(p \rightarrow q)$, $(q \rightarrow r)$ and $(r \rightarrow p)$ is True.

Case 1: We will prove $p \rightarrow q$ is True by contraposition.

$\therefore$ We will prove $\neg q \rightarrow \neg p$ is True.

$\neg p: 3x+2$ is not even, i.e. $3x+2$ is odd (By Defn. 1)

$\neg q: x+5$ is not odd, i.e. $x+5$ is even (By Defn. 1)

Let $\neg q$ be True. $\therefore (x+5)$ is even. By Defn 1, $\exists$ integer K s.t. $(x+5) = 2K$

$\therefore x = 2K - 5$. So, $3x+2 = 3(2K-5)+2 = 6K-15+2 = 6K-13 = (6K-14)+1$

$= 2(3K-7)+1$. It's easy to see that $3K-7$ is an integer. By Defn 1, $(3x+2)$ is an odd integer. $\therefore \neg p$ is True. By defn. of conditional, $\neg q \rightarrow \neg p$ is True. $\therefore (p \rightarrow q)$ is True.

Case 2: We will prove $q \rightarrow r$ is True by a direct proof

Let q be True. $\therefore x+5$ is odd. By Defn. 1, $\exists$ integer K such that $(x+5) = 2K+1$

$\therefore x = 2K - 4 = 2(K-2)$ $x^2 = (2(K-2))^2 = 4(K^2 - 4K + 4)$

$\Rightarrow x^2 = 2(2K^2 - 8K + 8)$. It's easy to see that $2K^2 - 8K + 8$ is an integer. By Defn 1, x^2 is even. $\therefore r$ is True. By defn. of conditional, $(q \rightarrow r)$ is True.

Case 3: We will prove $r \rightarrow p$ is True by contradiction

$\therefore$ We will prove $(r \wedge \neg p) \rightarrow \text{False}$ is True.

$\neg p: 3x+2$ is not even, i.e. $3x+2$ is odd.

Let $(r \wedge \neg p)$ be True. By defn. of AND, r is True and $\neg p$ is True.

$\therefore x^2$ is even and $3x+2$ is odd. By Defn 1, $\exists$ integers K_1 and K_2 such that $x^2 = 2K_1$ and $3x+2 = 2K_2+1$

$$\therefore x^2(3x+2) = 2K_1(2K_2+1) = (2K_1+2K_2)+1 = 2(K_1+K_2)+1$$